SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CSA1737 - Artifical Intelligence

CO1 - AT3

K. Subrahmanyam
192312521

1) Explain simple reflex, model-based, goal-based and utility based agents, analyze how each functions in a smart home, and justify which agent type or hybrid approach is most effective for comfort, energy efficiency and security.

Introduction:-

A Smart home automation system uses artificial intelligence (AI) to control lighting, temperature, and security based on user behaviour and environmental conditions.

It receives real-time inputs from motion sensors, temperature sensors, security devices and clocks.

An intelligent agent Perceives its environment through sensors and takes actions through actuators to achieve its objective. In a smart home, sensors Provide information about occupancy, temperature, and security.

Simple reflex agent:

A simple reflex agent selects actions based only on the Percepts, using condition-action rules. It does not maintain an internal model of Previous states.

Working in a smart home:-

=> If motion is detected in a room, switch on the lights.

=> If the room temperature exceeds a set threshold, switch on the cooling system.

Advantages:-

* Simple to design and implement.
* Responds quickly to real-time inputs.
* Suitable for repetitive tasks with clear rules.

Limitations:-

* Does not member Previous events.
* Cannot effectively handle situations requiring historical information.

Model based agent:

⇒ A model based agent maintains an internal state representing relevant information about the environment. it updates this state using current Percepts and Previous information.

Working in a smart home:

* Maintains the estimated occupancy status of different rooms.
* Tracks whether lights or appliances are already switched on.

Advantages:

* uses information from Previous observations.
* Handles Partially observable situations better than a simple reflex agent.

Limitations:-

* Requires memory and an internal representation of the home.
* Incorrect sensor information can produce an inaccurate internal state.

Goal based agent:-

A goal-based agent selects actions according to whether they help achieve a specified goal. It may consider future consequences and compare different possible actions.

Working in a smart home:-

The system may have goals such as:

* Maintain a comfortable indoor temperature.
* Ensure that doors are locked when occupants leave.
* Turn off unnecessary lights when rooms unoccupied.

Advantages:-

* Make decisions according to desired outcomes.
* Can plan a sequence of actions.
* Adapts its decisions when goals or conditions

Limitations:-

* Requires explict goals.
* Planning may require additional computation.

Utility based agents.

A utility-based agent evaluates Possibles actions using a utility function that represents how desirable each outcome is. It can compare alternatives when several objectives compete.

working in a smart home:-

* User comfort.
* Energy consumption and electricity cost.
* Security and safety.
* User Preferences.
* Device availability and operating conditions.

Advantages:-

* Balances competing objectives.
* Supports decisions under uncertainty.
* Can select an appropriate compromise between comfort, cost and security.

Limitations:-

* Designing an appropriate utility function can be difficult.
* Incorrect weights may Prioritize energy savings over comfort or security.

Most effective approach: Hybrid intelligent agents.

A hybrid architecture combining model-based, goal-based, utility-based decision-making, with simple reflex rules for urgent responses, is the most effective choice for the smart home described in the question.

Its operation can be organized as follows.

=> Perception: sensors collect information about motion, Temperature, time and security.

=> State update: The model-based component maintains the estimated occupancy.

=> Goal selection: The goal-based component identifies objectives such as comfort.

=> Action evaluation: The utility-based component compares available actions according to comfort.

=> immediate response: Simple Reflex rules handle urgent events, such as triggering -

(2) Explain how uniform cost search (UCS) and iterative Deepening search (IDS) can help a robot find an exit in an unknown maze. Compare their time and space complexities, relate them to intelligent agents, and justify the most effective combination.

Introduction:-

A robot Navigating an unknown maze must find a path from its starting Position to an exit without knowing the complete maze layout in advance. It must explore Possible Paths, avoid obstacles and determine when it has reached the exit.

Artifical Intelligence uses search algorithms to explore Possible states and identify a Path to a goal. Two Uniformed search strategies are uniform cost search (UCS) and iterative Deepening search.

Uniform cost search (UCS):

Uniform cost search expands the node with the lowest accumulated path g(n). It uses a Priority queue to select the least cost node.

Working Procedure:-

=> Place the robot's starting Position in a Priority queue with cost zero.

=> Remove the node with the lowest accumulated cost

→ Check whether the node represents the exit.
→ If it is not the exit, generate its reachable neighboring Positions.

Advantages:-

* Finds an optimal least cost path when all step costs are non-negative.
* Handles different movement costs.
* Is complete under the standard finite-branching Positive cost assumptions.

Disadvantages:-

* May explore many nodes before reaching the exit.
* can consume substantial memory because it stores frontier nodes.
* May be less efficient than depth-oriented methods when all steps have equal cost and memory is limited.

limited Deepening search (IDS)

Iterative Deepening search repeatedly Performs depth-limited depth-first search, increasing the depth limit after each unsuccessful iteration.

working Procedure:-

* set the initial depth limit to zero.
* start a depth-limited search from the robot's

starting position.
=> If the exit is found within the current limit, return the path.
=> otherwise, increase the depth limit by one.
=> Restart the search from the begining with the new limit.

Advantages:-
* uses much less memory than breadth-oriented searches.
* Is complete when the branching factor is finite.
* Finds a shortest path in terms of the number of edges when every step has equal cost.

Disadvantages:-
* Repeats the exploration of shallow nodes in successive iterations.
* Does not gurantee the best-cost path when step costs differ.

Relationship to intelligent agents:
The robot can be designed using different types of intelligent agents.

simple reflex agent: uses condition action rules based on the current percepts. For example, if the cell ahead is blocked, turn in another direction

Model-based agent: Maintains an internal map of explored cells and remembers previously visited positions.

Goal-based agent: uses the exit as its goal and selects actions that help reach it. It can use UCS or IDS to plan a route from the currently known maze.

Utility based agent: Evaluates outcomes according to factors such as travel cost, safety, time.

How agent design affects decision making

The choice of agent architecture determines how the robot uses information and select options.

A simple reflex agent reacts immediately but may repeatedly but may renters dead ends.

Most effective combination and justification.

For the maze described in the question, a model-based, goal-based agent combined with IDS is a good choice if every movement has equal cost and memory is limited.

The model-based component stores information about explored passages and obstacles.

=> The goal-based component defines the exit as the target state.

=> IDS explores paths with progressively increasing depth limits.

=> Its low memory requirement makes it suitable for a robot with limited computing resources..